

DPP-ADAPT

A method for **Architecture-Driven Adoption** of
Digital Product Passport-enabled **Procurement Transition**

Master's thesis — Presentation and First Evaluation of a Design Science Research artifact
Public Sector Innovation and eGovernance M. Sc. (PIONEER)
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Expert Evaluation
Questionnaire



<https://forms.cloud.microsoft/e/uLjYfAX0wu>



KU LEUVEN



Expert Evaluation: Friendly Instructions

- 1** Check this method presentation and DPP-ADAPT
Skim for relevance, completeness, consistency and fit to the problem dimensions.
- 2** Open the Excel workbook
Look through P0 (readiness), P2 (attributes), P4 (patterns) especially for authority context.
- 3** Walk a use case through the views
You can use the notebook example or select a suitable product from your context.
- 4** Think about a few criteria
→ Relevance, completeness, internal consistency, usability, fidelity, transferability.
- 5** Consider possible deal-breakers
What would block adoption in your opinion? Where does the method underestimate complexity?

*We will go through the presentation again during the evaluation interview.
In addition – or if you cannot find the time for an interview – please access the QR code/hyperlink on the first slide.*

Thank you very much for your support.

Public procurement is being asked to use a data infrastructure it cannot read (yet).

Literature can show that three main factors actively influence the current challenge.

1

ESPR & DPP

The EU Ecodesign for Sustainable Products Regulation (ESPR 2024/1781) introduces Digital Product Passports as “regulated, machine-readable product data infrastructure”.

2

Strategic procurement

Public buyers are increasingly expected to drive sustainable procurement through specifications, qualification, award criteria, and contract management.

3

Integration gap

Authorities lack governance, semantic mapping, evidence logic, and integration patterns to translate DPP data into procurement-defensible decisions.

The major challenge is a missing decision architecture.

Interviews show that the challenge is not purely technical.

Public procurement organisations lack an **enterprise-architecture-based integration logic** for integrating externally governed DPP data into procurement-relevant, reliable, and legally defensible decisions.

Source: literature synthesis triangulated with 11 experts through video interviews.

Reframing

DPP-procurement integration is not currently an interface challenge but can be better understood as an EA problem.

→ Solving it requires governance, semantics mapping, technical pattern selection beyond connecting an API.

Why a method, not a reference architecture or a checklist.

Three artifacts were considered.

Reference architecture

Prescribes a target state that low-maturity authorities **cannot reach** without substantial preparation.

not
preferred

Procedural checklist

Underestimates the architectural and governance dimensions; suitable only for narrow tasks.

not
preferred

Method (DPP-ADAPT)

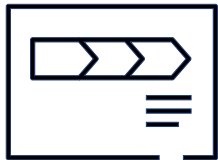
Combines architectural scope with applicable decision instruments/tools. Anchored to TOGAF ADM.

meets
expectations

Design Science Research + Method Engineering (Peffers et al., 2007; Brinkkemper, 1996; Henderson-Sellers & Ralyté, 2010)

DPP-ADAPT consists of two combined components + optional reference views.

Method and workbook are used together.



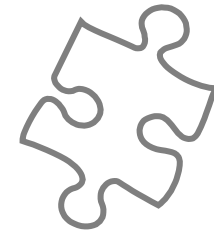
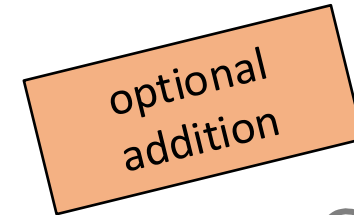
Method specification

This presentation + thesis document.
Defines 7 phases (P0–P6), six method components per phase, decision criteria, validity rules.



Workbook

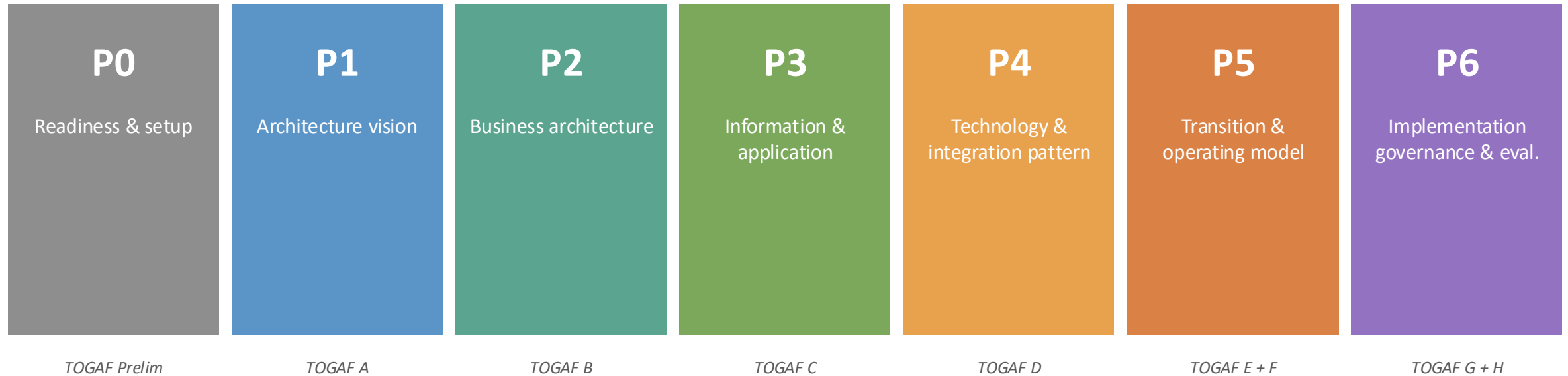
13 sheets in Excel: dashboard, instructions, ten phase sheets, reference data. Scoring + sign-off.



Architectural views

Three views in ArchiMate and BPMN, each mapped to a specific phase.

There are seven phases designed around TOGAF's ADM, adapted for DPP-PP.



Each phase is specified along six method components:

inputs, activities, techniques, tools, roles and outputs including decision criteria and validity rules.

...and few architectural views to describe key phases.



→ Available views can be accessed here: [DPP-ADAPT arch views](#)

Purpose: Assess readiness + set governance and decision rights.

Inputs

- Procurement & sustainability strategy
- Application inventory
- Sponsor candidates



Activities

- 25-statement readiness self-assessment (5 dimensions × 5 statements)
- Stakeholder register & RACI
- Preliminary operating-model choice



Outputs

- Readiness level (Low/Medium/High)
- Approved RACI
- Sponsor sign-off

In the workbook:

Sheets 3 (Readiness) and 4 (RACI). The overall score is auto-computed and traffic light colored (red/orange/green).

Purpose: Confirm motivation, choose product family, define in-scope process stages

Inputs

- P0 outputs
- Sustainability objectives
- Recurring procurements



Activities

- Drivers and goals
- ONE product family selection
- In-scope stage selection
- Capability gap



Outputs

- Scope canvas
- Capability gap analysis

Decision criterion:

Exactly one product family in scope for first implementation. If governance dimension low, plan P6 in parallel with P5.

In the workbook:

Sheet 5: Y/N drop-downs for stages, 1–5 scoring for drivers and capability gap.

Purpose: Select touchpoints, score DPP attributes, decide procurement-use roles.

Inputs

- P1 outputs
- DPP attribute candidates (illustrative)
- Procurement-law guidance



Activities

- Touchpoint matrix
- Attribute scoring on relevance (R), evidence confidence (E), and verifiability (V) on 1–5
- Procurement-use role decisions



Outputs

- Approved attributes by use-role
- View R1 (BPMN)
- Risk-gate outcome

In the workbook:

Sheet 7: incl. each attribute's scale and corresponding item (EU energy label, ESPR, REACH, etc.).

Purpose: Semantic mapping, information lifecycle, application inventory

Inputs

- P2 outputs
- Application inventory
- Semantic references



Activities

- Per-attribute semantic mapping with type, rule, unit alignment, sign-off
- Information lifecycle (retention ≥ 60 mo)
- Apps in R2



Outputs

- Approved mapping table
- Information lifecycle
- View R2

In the workbook:

Sheet 8: drop-downs for mapping type, status, unit alignment. Sheet 9 — applications and retention check.

Purpose: Score patterns A–E; document rationale.

Inputs

- P0 readiness
- P2 touchpoints
- P3 mapping



Activities

- Score each pattern on 7 dimensions
- Calculate weighted total
- Document reasoning statement



Outputs

- Chosen pattern (A/B/C)
- Weighted total scoring

In the workbook:

Sheet 10: pattern scoring with weight-sum check. Result auto-pulled into the dashboard.

Purpose: Plateau roadmap, work packages, risks and risk management

Inputs

- P0–P4 outputs
- Portfolio context



Activities

- Three plateaus (current/first/scaled)
- Work-package register
- Risk register with RAG



Outputs

- Roadmap
- Work packages with dependencies
- Risk register

Decision criterion:

Plateau 1 reachable within around 12 months by default.

In the workbook:

Sheet 11: M = months from project start, PD = person-days, WP = work package. Drop-downs for WP status and plateau.

Purpose: Governance, indicators, expert review

Inputs

- P5 outputs
- Existing governance bodies



Activities

- Governance charter
- Indicator catalogue
- Ex-ante expert review



Outputs

- Complete Charter
- Indicators with targets
- Expert review report

Decision criterion:

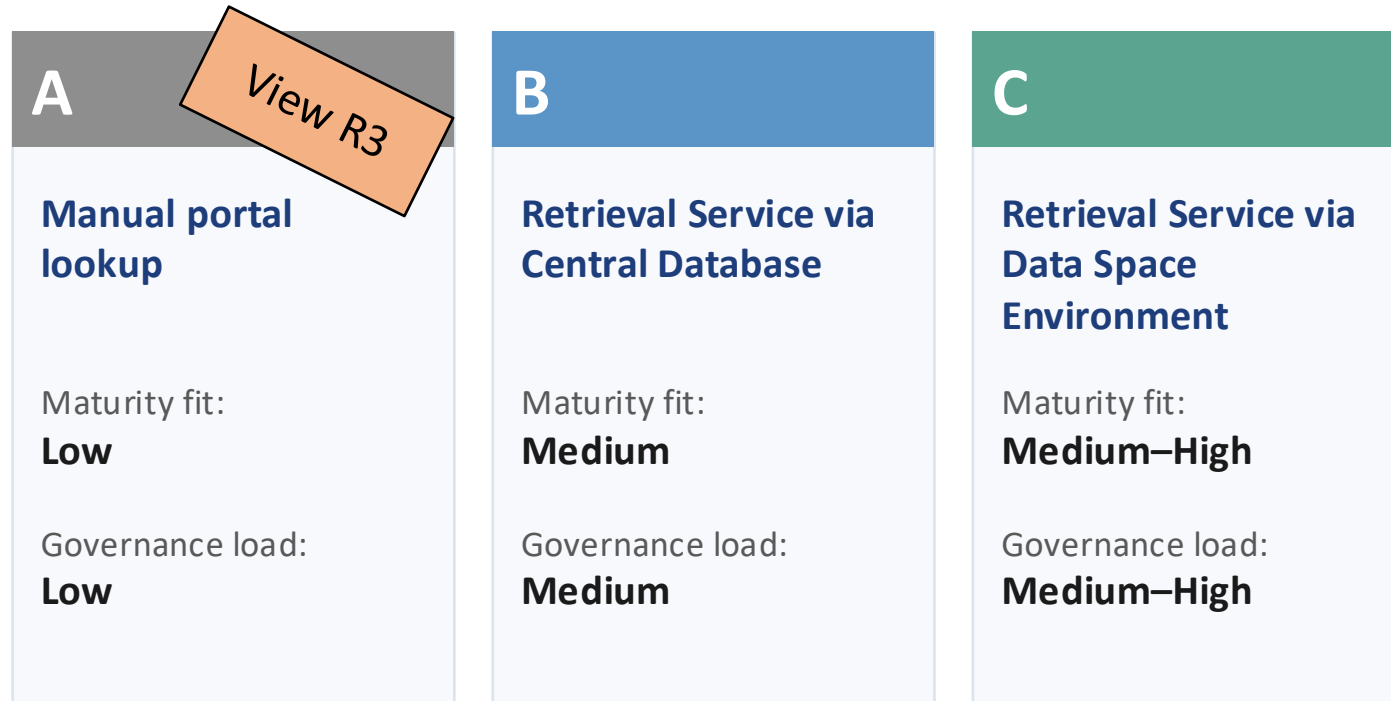
Panel ≥ 3 experts from PP/EA/sustainability/legal. Criteria avg. < 3.0 triggers improvement before broader roll-out.

In the workbook:

Sheet 12: expert ratings 1–5, indicators with target vs. current and last-review date.

Three integration patterns to choose from according to maturity level.

Each approach comes with a different level of integration across technology in use.



Pattern selection is scored on 7 dimensions (Sheet 10): readiness fit, use-case fit, scalability, governance load, cost, time-to-value, future-proofness.

Default weights and rationale are on Sheet 2.

Worked example: laptop procurement

Fraunhofer ISST has been chose to demonstrate the method in a working environment.

→ See workbook for full example (Dashboard)

Fraunhofer ISST, headquartered in Dortmund, Germany, has approx. 120 notebook users on a 3 to 4 year refresh (€200–400K/cycle) running on SAP S/4HANA + SAP Ariba via central Fraunhofer ICT and Microsoft Entra ID, combined with in-house dataspace expertise and existing Kubernetes capability.

Fraunhofer ISST	
Staff	191
Operating budget	€ 12.25 M / yr
Notebook users	approx. 120
Refresh cycle	3–4 years
Cycle cost	€ 200–400 K
e-Procurement	SAP Ariba
ERP	SAP S/4HANA
IDP	MS Entra ID
Directive	2014/24/EU
Readiness 3.88 → HIGH	
Chosen Pattern C	



Outcome: Pattern C selected (procurement-platform retrieval service via Data Spaces). Two criteria approved (energy efficiency, repairability). 12-month plateau plan to first DPP-informed tender.

Workbook contains 12 sheets, every output is a single checkpoint.

1 Dashboard

Steering-committee view + charts

2 Method map

Weights and rationale, version log

3 P0 Readiness

25 statements × 5 dimensions

4 P0 RACI

Stakeholders + decision rights

5 P1 Scope

Drivers, goals, product family

6 P2 Touchpoints

Stage × use-role matrix

7 P2 Attributes

Scale, meaning, source, score

8 P3 Mapping

Semantic mapping & sign-off

9 P3 Apps

Application & info lifecycle

10 P4 Pattern

7-dimension scoring & choice

11 P5 Roadmap

Plateaus, WPs (M, PD), risks

12 P6 Evaluation

Indicators & expert ratings

Thank you for your time. Please fill in the questionnaire.

Thank you very much for your support.

Expert Evaluation Questionnaire



<https://forms.cloud.microsoft/e/uLjYfAX0wu>